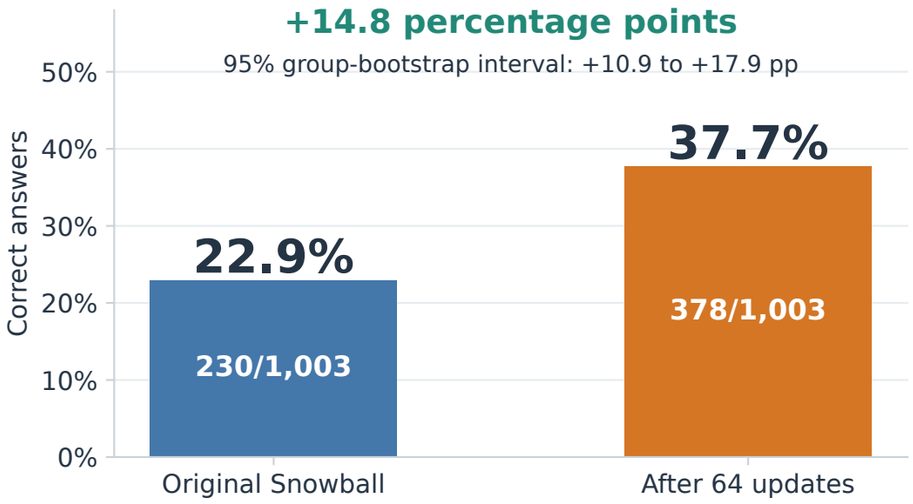
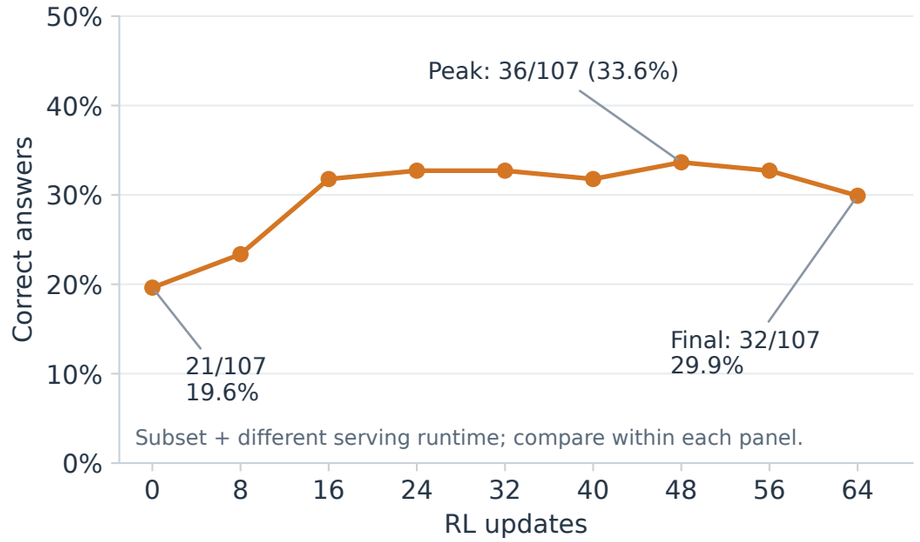
Snowball RL pilot: a real score gain, with important limits

64 RL updates • preselected final checkpoint • no teacher traces

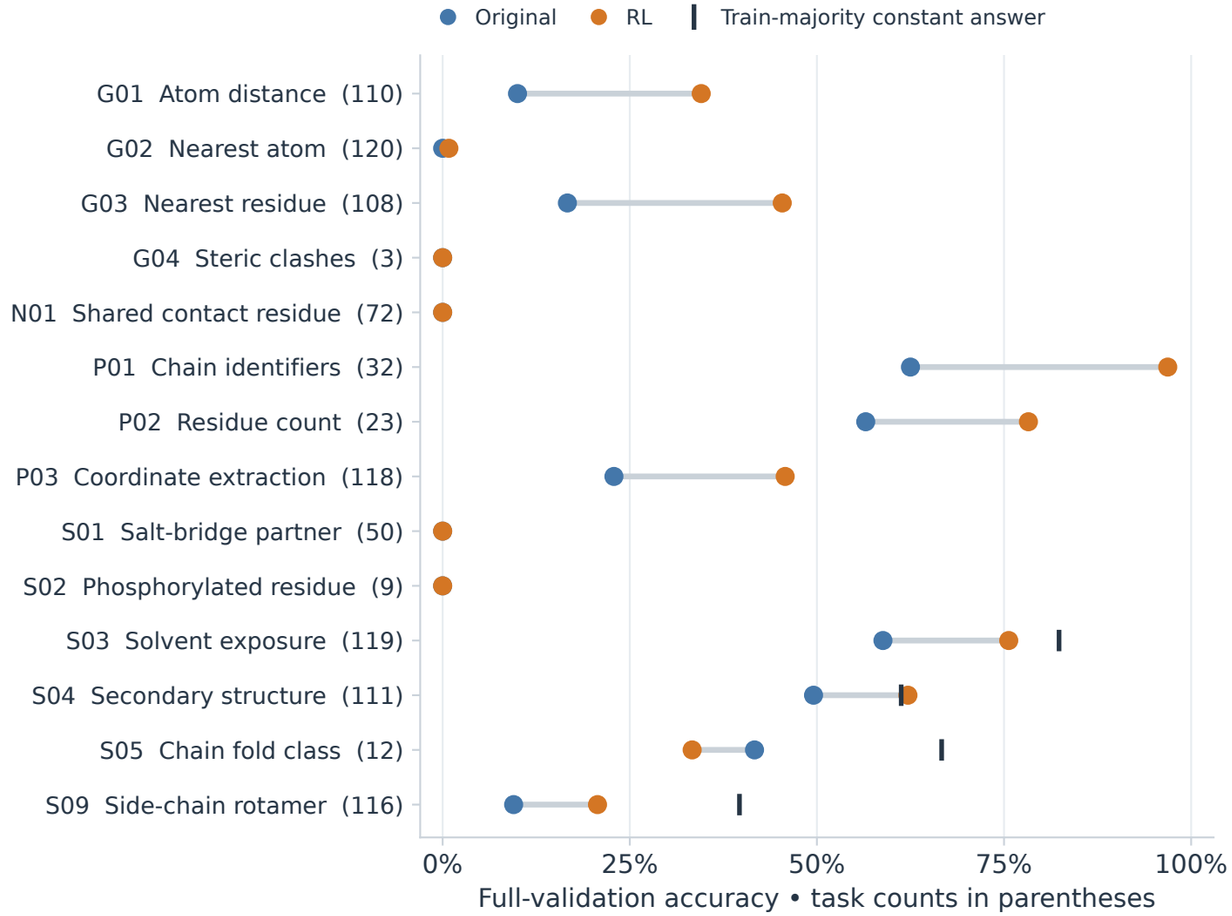
A Matched full validation



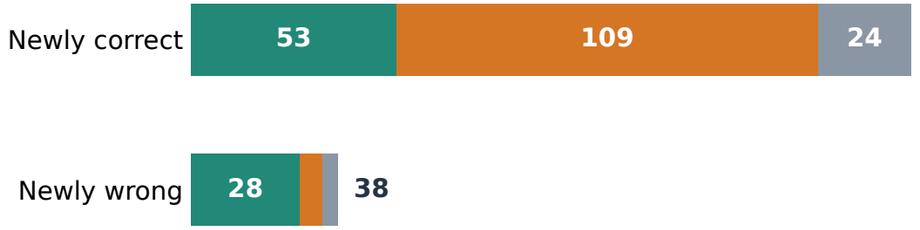
B Learning curve: fixed 107-task monitor



C Gains by task family



D What changed in the paired answers?



133/186 gains (72%) began with an invalid response.

Valid before and after (412 tasks):
53 gains − 28 losses = +25 correct

Descriptive subset; not a causal formatting ablation.
*Truncation/refusal/tool failure without a format-error flag.

Evidence: 1,003 paired validation tasks from 20 held-out source groups; 19/20 groups improved.
All 2,006 scores replay exactly. Approximate 95% interval: 50,000 paired group resamples (seed 17).
One RL run; no seed-replication interval. See the paired-test comparison for the separate 1,370-task result.